

An Investigation of a Notion of „Variable Forgetting“ that Minimizes Truth Values

Bachelor's Thesis

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submitted by
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Abstract

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1 Motivation

Propositional logic deals, among other things, with the satisfiability of sentences or the inference of sentences from a set of premises. A knowledge base may contain many different statements in various respects. To assert or entail sentences about specific phenomena or interest, it would be advantageous to work only with relevant knowledge to reduce computational complexity. In other words, non-relevant or irrelevant information in relation to the current interest could be "forgotten". Furthermore, knowledge bases change over time, and old knowledge might need to be contracted. To this end, an operation exists for eliminating variables within a proposition. As stated in [LR94] or [LLM03], we define variable forgetting inductively as follows:

Definition 1 (Skeptical Variable Forgetting) *Let F be a formula, p an atomic variable and S a Set of Variables. The skeptical forgetting of p in F $SkepForget(F, p)$, respectively, the skeptical forgetting of S in F $SkepForget(F, S)$, is given inductively by:*

- $SkepForget(F, \emptyset) = F$
- $SkepForget(F, \{p\}) = F[p/\top] \wedge F[p/\perp]$
- $SkepForget(F, p) = SkepForget(F, \{p\})$
- $SkepForget(F, S \cup \{p\}) = SkepForget(SkepForget(F, \{p\}), S)$

For some formulas F , this operation of variable elimination can lead to a notable reductive or drastic results. We start by considering an example where *Forget* behaves quite appropriate. Let a sentence F_1 be $a \wedge b$ and we choose to forget about b :

$$\begin{aligned} Forget(F_1, b) &= F_1[b/\top] \vee F_1[b/\perp] \\ &\equiv ((a \wedge \top) \vee (a \wedge \perp)) \\ &\equiv (a \vee \perp) \\ &\equiv a \end{aligned}$$

We can see, as shown above, that $Forget(F_1, b)$ is equivalent to a which seems to be the intuitive result when thinking of forgetting about b in F_1 . Next, we consider forgetting of b in the formula $F_2 = a \vee b$

$$\begin{aligned} Forget(F_2, b) &= F_2[b/\top] \vee F_2[b/\perp] \\ &\equiv ((a \vee \top) \vee (a \vee \perp)) \\ &\equiv (\top \vee a) \\ &\equiv \top \end{aligned}$$

We observe that $Forget(F_2, b)$ is *tautological*, which can be seen as a rather drastic result when we choose to forget b only. A variation of a forget operation could be defined as follows:

Definition 2 (Skeptical Variable Forgetting) Let F be a formula, p an atomic variable and S as Set of Variables. The skeptical forgetting of p in F $SkepForget(F, p)$, respectively, the skeptical forgetting of S in F $SkepForget(F, S)$, is given inductively by:

- $SkepForget(F, \emptyset) = F$
- $SkepForget(F, \{p\}) = F[p/\top] \wedge F[p/\perp]$
- $SkepForget(F, p) = SkepForget(F, \{p\})$
- $SkepForget(F, S \cup \{p\}) = SkepForget(SkepForget(F, \{p\}), S)$

The idea is to use conjunction instead of disjunction. The consequences for our previous examples are the following:

$$\begin{aligned} SkepForget(F_1, b) &= F_1[b/\top] \vee F_1[b/\perp] \\ &\equiv ((a \wedge \top) \wedge (a \wedge \perp)) \\ &\equiv (a \wedge \perp) \\ &\equiv \perp \end{aligned}$$

$$\begin{aligned} SkepForget(F_2, b) &= F_2[b/\top] \vee F_2[b/\perp] \\ &\equiv ((a \vee \top) \wedge (a \vee \perp)) \\ &\equiv (\top \wedge a) \\ &\equiv a \end{aligned}$$

We observe that our alternative operation, at least in these examples, doesn't necessarily entail less drastic results. We obtain a contradiction in the first case and in the second case, our equivalent to a . It suggests a duality between the two operations.

2 SyntacticAnalysis

In the previous section we gave examples of sentences with only one connective. But what about more complex sentences. Does the outcome predominantly correlate with the top layer of connectives in the syntax tree? What is the effect or influence of deeper nested expressions on the final result? To take the investigation a step further, and also for the semantic analysis, we need to identify examples that are at least general enough to make some valuable statements. But how to find them, what is important? First let's experiment with some more sentence and also deeper nesting and compare the results!

$F_1 \equiv a$	$Forget(F_1, a) \equiv \top$	$SkepForget(F_1, a) \equiv \perp$
$F_{1b} \equiv \neg a$	$Forget(F_{1b}, a) \equiv \top$	$SkepForget(F_{1b}, a) \equiv \perp$
$F_2 \equiv (a \wedge b)$	$Forget(F_2, a) \equiv b$	$SkepForget(F_2, a) \equiv \perp$
$F_{2b} \equiv \neg(a \wedge b)$	$Forget(F_{2b}, a) \equiv ((\perp \vee \neg b) \vee (\top \vee \neg b))$	$SkepForget(F_{2b}, a) \equiv ((\perp \vee \neg b) \wedge (\top \vee \neg b))$
$F_3 \equiv (a \wedge (c \wedge d))$	$Forget(F_3, a) \equiv (c \wedge d)$	$SkepForget(F_3, a) \equiv \perp$
$F_5 \equiv (a \vee b)$	$Forget(F_5, a) \equiv \top$	$SkepForget(F_5, a) \equiv b$
$F_6 \equiv (a \vee (c \vee d))$	$Forget(F_6, a) \equiv \top$	$SkepForget(F_6, a) \equiv (c \vee d)$
$F_7 \equiv (a \vee (c \wedge d))$	$Forget(F_7, a) \equiv \top$	$SkepForget(F_7, a) \equiv (c \wedge d)$
$F_8 \equiv (a \wedge (c \vee d))$	$Forget(F_8, a) \equiv (c \vee d)$	$SkepForget(F_8, a) \equiv \perp$

Now here we can see... this and that

2.1 Behavior of Truth Values in Abstract Syntax Trees

The behavior of truth values presents notably insights when considering their interactions with logical connectives. We know that $\top \wedge \perp \equiv \perp$ and $\top \vee \perp \equiv \top$.

To investigate the behaviour abstract syntax trees are a great way to visualize sentences. To get a good overview we will try any permutation of connectives over a certain variable amount n which is $2^{(n-1)}$ possibilities.

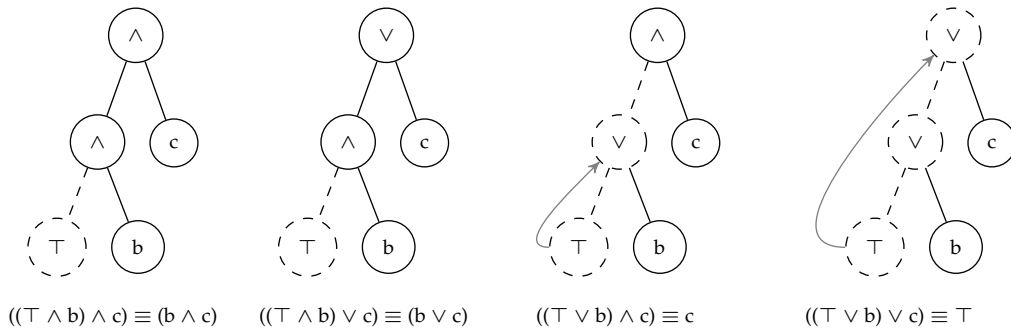


Figure 1: $\top$ propagation

In figure 3 we can observe the propagation of truth values, in this case $\top$. Starting from the leafes $\top$ effectively "absorbs" all siblings in conjunction with $\vee$ and stops propagating at $\wedge$.

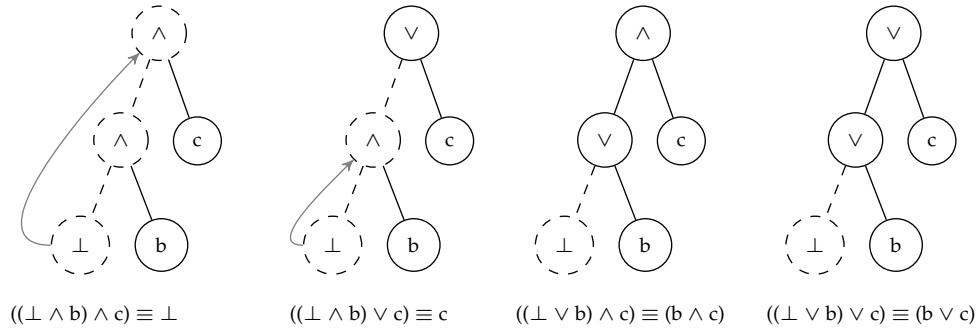
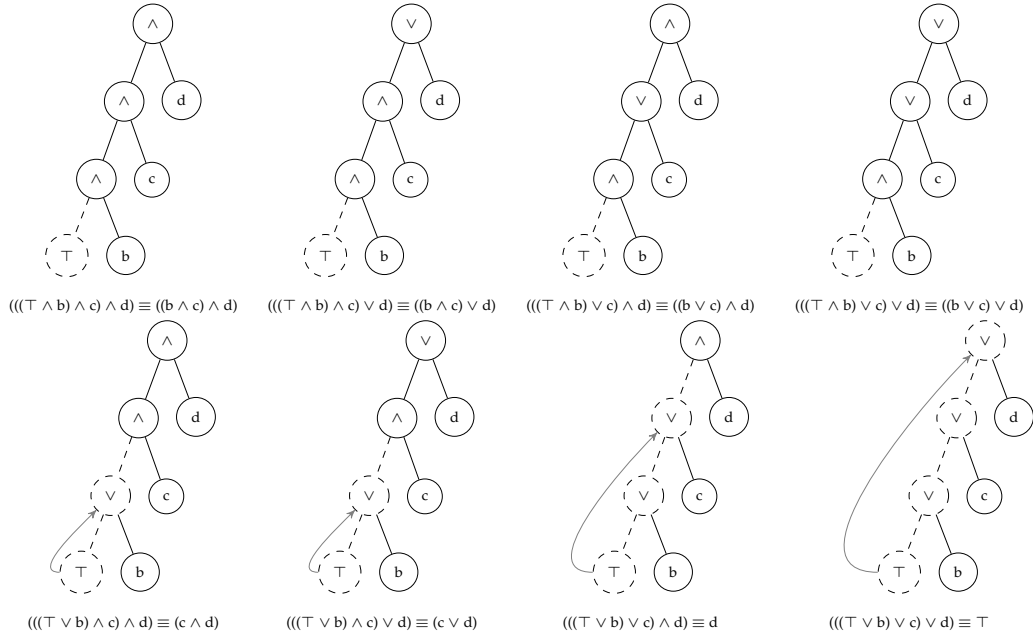


Figure 2: $\perp$ propagation

Figure 2 shows, dual to 3, how $\perp$ "annihilates" all siblings when paired with a logical $\wedge$ and halts at $\vee$. This dynamic interaction implies that truth values possess the capability to selectively "eliminate" variables and entire propositions.



3 SemanticInvestigation

A	B	C	$((A \vee B) \wedge C)$	$((T \vee B) \wedge C) \vee ((\perp \vee B) \wedge C) \equiv C$
0	1	1	1	1
0	1	0	0	0
0	0	1	0	1
0	0	0	0	0
1	1	1	1	1
1	1	0	0	0
1	0	1	1	1
1	0	0	0	0

Figure 4: Forget $((A \vee B) \wedge C, A)$

A	B	C	$((A \vee B) \wedge C)$	$((T \vee B) \wedge C) \wedge ((\perp \vee B) \wedge C) \equiv (B \wedge C)$
0	1	1	1	1
0	1	0	0	0
0	0	1	0	0
0	0	0	0	0
1	1	1	1	1
1	1	0	0	0
1	0	1	1	0
1	0	0	0	0

Figure 5: SkipForget $((A \vee B) \wedge C, A)$

A	B	C	$((A \wedge B) \vee C)$	$((T \wedge B) \vee C) \vee ((\perp \wedge B) \vee C) \equiv (B \vee C)$
0	1	1	1	1
0	1	0	0	1
0	0	1	1	1
0	0	0	0	0
1	1	1	1	1
1	1	0	1	1
1	0	1	1	1
1	0	0	0	0

Figure 6: Forget $((A \wedge B) \vee C, A)$

A	B	C	$((A \wedge B) \vee C)$	$((T \wedge B) \vee C) \wedge ((\perp \wedge B) \vee C) \equiv C$
0	1	1	1	1
0	1	0	0	0
0	0	1	1	1
0	0	0	0	0
1	1	1	1	1
1	1	0	1	0
1	0	1	1	1
1	0	0	0	0

Figure 7: SkipForget $((A \wedge B) \vee C, A)$

A	B	C	$((A \vee B) \wedge C)$	$((A \vee B) \wedge \top) \vee ((A \vee B) \wedge \perp) \equiv (A \vee B)$
1	1	0	0	1
1	0	0	0	1
0	1	0	0	1
0	0	0	0	0
1	1	1	1	1
1	0	1	1	1
0	1	1	1	1
0	0	1	0	0

Figure 8: $\text{Forget}((A \vee B) \wedge C, C)$

A	B	C	$((A \vee B) \wedge C)$	$((A \vee B) \wedge \top) \wedge ((A \vee B) \wedge \perp) \equiv \perp$
1	1	0	0	0
1	0	0	0	0
0	1	0	0	0
0	0	0	0	0
1	1	1	1	0
1	0	1	1	0
0	1	1	1	0
0	0	1	0	0

Figure 9: $\text{SkepForget}((A \vee B) \wedge C, C)$

A	B	C	$((A \wedge B) \vee C)$	$((A \wedge B) \vee \top) \vee ((A \wedge B) \vee \perp) \equiv \top$
1	1	0	1	1
1	0	0	0	1
0	1	0	0	1
0	0	0	0	1
1	1	1	1	1
1	0	1	1	1
0	1	1	1	1
0	0	1	1	1

Figure 10: $\text{Forget}((A \wedge B) \vee C, C)$

A	B	C	$((A \wedge B) \vee C)$	$((A \wedge B) \vee \top) \wedge ((A \wedge B) \vee \perp) \equiv (A \wedge B)$
1	1	0	1	1
1	0	0	0	0
0	1	0	0	0
0	0	0	0	0
1	1	1	1	1
1	0	1	1	0
0	1	1	1	0
0	0	1	1	0

Figure 11: $\text{SkepForget}((A \wedge B) \vee C, C)$

$$\text{Mod}(\text{Forget}(F, x)) = \text{Mod}(F) \cup \{\text{Switch}(w, x) \mid w \models F\}$$

$$\text{Mod}(\text{SkepForget}(F, x)) = \text{Mod}(F) \cap \{\text{Switch}(w, x) \mid w \models F\}$$

References

- [LLM03] Jérôme Lang, Paolo Liberatore, and Pierre Marquis. Propositional independence-formula-variable independence and forgetting. *Journal of Artificial Intelligence Research*, 18:391–443, 2003.
- [LR94] Fangzhen Lin and Ray Reiter. Forget it. In *Working Notes of AAAI Fall Symposium on Relevance*, pages 154–159, 1994.